Work Plan

Spectral Methods for Binary Neural Networks

Addressing Committee Review & Research Milestones

February 2026

[REVIEW]	Task directly addresses committee feedback
[THESIS]	Task from the proposed research program
[BOTH]	Addresses review *and* advances the research

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1 Executive Summary

The committee's review identified five major and four moderate concerns. The single most critical action item is the **pilot spectral concentration experiment** — the reviewer explicitly conditioned candidacy advancement on presenting this evidence. The second critical item is **at least one formal theoretical result**.

This work plan is organized into five phases, ordered so that each phase either (a) addresses review concerns that gate subsequent work, or (b) builds on infrastructure from the prior phase. The estimated total timeline is 10–12 months of focused work, with the pre-defense deliverables (Phases 1–2) completable in approximately 8–10 weeks.

Key principle. We front-load the highest-risk, most informative experiments. If the pilot study shows no spectral concentration (Phase 1), we pivot early rather than investing months in an approach built on a false assumption.

2 Phase 1: Pilot Study and Foundation (Weeks 1–4)

Goal: Test the core assumption and build infrastructure. This phase produces the evidence the committee requires before the defense.

2.1 Task 1.1: Implement spectral analysis tooling [BOTH]

What. Build the `PilotAnalysis` module:

- Fast Walsh–Hadamard transform for vectors of length 2^n (optimized NumPy/PyTorch implementation).
- Functions to compute: full Walsh–Fourier spectrum of a binary vector, spectral energy by degree $E(k) = \sum_{|S|=k} \hat{w}(S)^2$, effective spectral degree k^* at the 95% energy threshold, spectral entropy, and the per-row influence profile.
- Visualization: spectral energy heatmaps (layer $\times$ degree), histograms of k^* across rows, comparison against random baselines.

Estimated effort. 3–4 days. The FWHT is $\sim$20 lines of code; the analysis functions are straightforward.

Addresses. Major Concern 1 (pilot experiment), Suggestion 1 (run it now).

2.2 Task 1.2: Obtain and analyze binary weight matrices [BOTH]

What. Run the pilot study protocol described in the revised thesis (§5):

1. **Source A:** Download BitNet b1.58 2B4T open weights. Extract ternary weight matrices. Analyze as pseudo-Boolean functions.
2. **Source B:** Load GPT-2 Small (124M). Binarize all weight matrices via $\mathrm{sgn}(W)$. Analyze.
3. **Source C:** (Deferred to Task 2.2 — requires training a binary model first.)
4. **Controls:** Random ± 1 matrices of matched dimensions; random matrices with matched row-sum distribution; random linear threshold functions (LTFs) generated by $w(j) = \mathrm{sgn}(\langle a, \mathrm{bin}(j)\rangle)$ with $a \sim \mathcal{N}(0, I_n)$ where $n = \log_2 d$. The LTF control is critical: if trained weights look spectrally similar to random LTFs, the observed concentration reflects sign-activation structure rather than task-specific learning.

For each matrix, compute the full spectral profile. Report the distribution of effective spectral degree k^* across rows and layers.

Non-power-of-two dimensions. Since BitNet b1.58 2B4T uses $d = 2560$ and GPT-2 Small uses $d = 768$, neither is a power of two. We handle this via (i) blockwise power-of-two submatrix analysis, and (ii) zero-padding to the next power of two, reporting both to quantify encoding artifacts.

Estimated effort. 2–3 days for Sources A and B plus controls. Estimated 1–6 CPU-hours depending on model size, threading, and permutation repeats.

Decision gate. If k^* is consistently close to n (no concentration) across all sources: **stop and reassess.** Possible pivots:

- Shift primary focus to Hadamard-structured layers (which do not depend on row-wise spectral sparsity).
- Investigate whether concentration emerges during spectral *training* even if absent in conventionally trained weights.
- Reframe the thesis contribution around the theoretical framework and diagnostic tools rather than performance claims.

Addresses. Major Concern 1 (the load-bearing assumption), Suggestion 1 (run the pilot now).

2.3 Task 1.3: Column-indexing sensitivity analysis [REVIEW]

What. Using the spectral data from Task 1.2, test whether the spectral profile is sensitive to column ordering:

1. Verify (as theory predicts) that the degree-level energy profile $\sum_{|S|=k} \hat{w}(S)^2$ is invariant under bit permutations.
2. Measure whether it changes significantly under 20 random column permutations.
3. Compare standard binary encoding vs. Gray code encoding.

Estimated effort. 1 day (reuses infrastructure from Task 1.1).

Addresses. Major Concern 2 (column-indexing arbitrariness), Suggestion 5.

2.4 Task 1.4: Toy model — Majority function [REVIEW]

What. Implement and run the toy example from the revised thesis (§4.8):

1. Build a minimal `SpectralLinear` layer for $d = 8$ ($n = 3$).
2. Train it to learn Maj_3 using spectral coefficients at degree $k = 1, 2, 3$.
3. Verify convergence to the known Fourier spectrum.
4. Extend to Maj_7 ($d = 128$, $n = 7$) and the tribes function.
5. Log all spectral diagnostics; visualize training dynamics.

Estimated effort. 2–3 days.

Addresses. Suggestion 6 (toy model), and serves as a sanity check for the `SpectralLinear` implementation before scaling.

2.5 Phase 1 Deliverables

1. Spectral concentration results for BitNet and binarized GPT-2 weight matrices (figures + summary statistics).
2. Column-indexing sensitivity report.
3. Working toy-model demonstration with training curves and spectral diagnostics.
4. Go/no-go decision on the row-wise spectral parameterization.

3 Phase 2: Theory and Core Implementation (Weeks 5–10)

Goal: Produce at least one formal theoretical result and build the core `SpectralLinear` *and* `HadamardLinear` *modules. Combined with Phase 1, this constitutes the defense-ready package.*

3.1 Task 2.1: Strengthen theoretical results [REVIEW]

The revised thesis already includes two preliminary results (the degree-k approximation bound and the normalization propagation proposition). The committee asks for "at least one formal theoretical result." We aim for three contributions, in decreasing order of confidence:

Result A (high confidence): Tighten the approximation bound. The current Theorem 3.1 in the revised thesis bounds the error of degree-k truncation in terms of total influence. Strengthen this by:

- Proving a tighter bound for the specific case where f is a linear threshold function, using quantitative noise sensitivity bounds for LTFs (e.g., from Harsha et al.).
- Deriving an explicit bound on the number of bit flips introduced by the sign projection as a function of k and the spectral gap.

Result B (medium confidence): Spectral-domain gradient characterization. Prove that the gradient of the cross-entropy loss with respect to spectral coefficients $\hat{w}(S)$ decomposes as:

$$\frac{\partial \mathcal{L}}{\partial \hat{w}_i(S)} = \sum_j \frac{\partial \mathcal{L}}{\partial w_i(j)} \cdot \chi_S(\mathrm{bin}(j)),$$

which is itself a Walsh–Fourier coefficient of the spatial gradient. Note that this identity alone is just a change of basis and does not imply any advantage. The formal goal is to show that *combined with degree truncation or L_1/degree-penalty regularization in the spectral domain*, gradient descent converges faster to low-complexity Boolean functions than weight-domain STE. This is a formal version of the "better inductive bias" claim; without the sparsity-inducing component, no separation should be expected.

Result C (aspirational): Separation result. Show that there exists a family of Boolean functions for which spectral parameterization at degree $k = O(\log n)$ converges to the optimum in polynomial time, while weight-domain STE requires exponential time. The candidate family: random k-juntas (functions depending on only k of n variables), for

which the Kushilevitz–Mansour algorithm gives polynomial-time learning but brute-force search over ± 1 weights is exponential.

Estimated effort. 2–3 weeks. Result A is a relatively straightforward extension of known bounds. Result B requires careful but standard calculus of the WHT. Result C is genuinely hard and may remain a conjecture.

Pivot plan for Result B. If a clean formal proof of the gradient advantage cannot be obtained within the allocated time, we pivot to a rigorous empirical result: measure gradient variance, spectral sparsity, and convergence speed in both spectral and weight domains across multiple random seeds, problem scales, and regularization strengths. Present the comparison with confidence intervals and statistical tests. This produces a "rigorous empirical result" that satisfies the committee's requirement for formal analysis even if a closed-form proof eludes us.

Addresses. Major Concern 5 (missing theory).

3.2 Task 2.2: Implement `SpectralLinear` [THESIS]

What. Production-quality PyTorch module:

- Parameters: spectral coefficient tensor of shape $[d_{\text{out}}, \binom{n}{\leq k}]$.
- Precomputed: Walsh basis matrix of shape $[\binom{n}{\leq k}, d_{\text{in}}]$ (rows of H_n corresponding to monomials of degree $\leq k$).
- Forward pass: $W_{\text{cont}} = \hat{w} \cdot \Phi^\top$; $W_{\text{bin}} = \tanh(\beta \cdot W_{\text{cont}})$ (or sgn with STE); output $= W_{\text{bin}} \cdot x$.
- Backward pass: autograd handles everything (STE through sign; exact through linear WHT map).
- Spectral norm projection as a post-step hook.

Estimated effort. 1 week.

Validation. Verify on the toy tasks from Task 1.4 that the module produces identical results to the prototype. Benchmark forward/backward speed against `nn.Linear`.

3.3 Task 2.3: Implement `HadamardLinear` [THESIS]

What. PyTorch module implementing the butterfly factorization:

- Parameters: L_H sign vectors $s_\ell \in \{-1, +1\}^d$; L_H soft permutation matrices (Sinkhorn-parameterized during training, hard at inference).
- Forward: cascade of FWHT $\to$ sign-flip $\to$ permute, with explicit $d^{-L_H/2}$ normalization.
- FWHT implementation using the standard $O(d \log d)$ butterfly algorithm.

Estimated effort. 1 week.

Addresses. Moderate Concern 7 (Sinkhorn leakage) — implement hard-rounding ablation from the start.

3.4 Task 2.4: Train a baseline binary Transformer (STE) [THESIS]

What. Train Baseline 2 (standard binary STE, no spectral parameterization) on WikiText-103 at $d = 256$, $L = 4$. This serves three purposes:

1. Provides the binary weight matrices needed for Source C of the pilot study (completing Task 1.2).
2. Establishes the baseline perplexity to beat.
3. Validates the training infrastructure (data loading, evaluation, logging) before adding spectral components.

Estimated effort. 3–5 days (training time dominates; implementation reuses standard Transformer code with sign quantization).

3.5 Phase 2 Deliverables

1. At least one (ideally two) formal theoretical results with complete proofs.
2. Working, tested `SpectralLinear` and `HadamardLinear` modules.
3. STE baseline results on WikiText-103 ($d = 256$, $L = 4$).
4. Completed pilot study including Source C (STE-trained binary weights).
5. **Defense-ready package:** revised thesis + pilot study results + theoretical results.

4 Phase 3: Core Experiments (Weeks 11–20)

Goal: Train and evaluate the Spectral Binary Transformer. Run the primary ablation studies.

4.1 Task 3.1: Implement the full SBT architecture [THESIS]

What. Assemble `SpectralTransformerBlock` from the modules built in Phase 2:

- `SpectralLinear` for Q, K, V, output projection, and FFN layers.
- Integer attention with fixed $1/\sqrt{d_k}$ scaling (Option A — the recommended starting point).
- RMSNorm or LayerNorm (at appropriate precision).
- Residual connections.
- Spectral diagnostics logging (effective degree, entropy, rounding error per layer per N steps).

Estimated effort. 1 week.

4.2 Task 3.2: Small-scale SBT training runs [THESIS]

What. Train the SBT at the smallest configuration ($d = 256$, $L = 4$, $k = 3$) on WikiText-103:

1. Establish that training converges (loss decreases; spectral diagnostics are sensible).
2. Compare against Baseline 2 (STE) and Baseline 4 (random frozen binary).
3. Sweep learning rate, temperature schedule β_t, and regularization strength λ.

Estimated effort. 1–2 weeks (many short training runs).

Decision gate. If SBT at $d = 256$ cannot outperform random binary weights after hyperparameter tuning: investigate whether the issue is (a) the spectral degree k, (b) the projection method, or (c) something more fundamental. Run ablations A1 and A2 at this scale before scaling up.

4.3 Task 3.3: Primary ablation studies [THESIS]

Run ablations A1–A6 at the small scale ($d = 256$, $L = 4$):

Ablation	Variable	Settings
A1	Spectral degree k	$1, 2, 3, 4, \ldots, 8$
A2	Projection method	Hard sign, randomized, spectral norm
A3	Structured vs. unstructured	SpectralLinear vs. HadamardLinear
A4	Spectral regularization	$\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}\}$; $\mu \in \{0, 10^{-3}\}$
A5	Activation precision	Binary / 8-bit / full precision
A6	Embedding strategy	Learned FP / learned binary / Hadamard

Estimated effort. 2–3 weeks (many runs, but each is fast at this scale).

Key outputs.

- Optimal k^* for the spectral degree (A1). This is the most important result.
- Whether projection method matters significantly (A2).
- Whether Hadamard structure helps or hurts (A3).
- Whether spectral regularization improves convergence (A4).

4.4 Task 3.4: Scale-up experiments [THESIS]

Using the best configuration from Task 3.3, train at larger scales:

1. $d = 512$, $L = 8$ on WikiText-103.
2. $d = 1024$, $L = 12$ on WikiText-103 (and/or Pile subset, if compute permits).
3. Compare against all four baselines at each scale.

Estimated effort. 2–3 weeks (longer training runs; fewer configurations).

4.5 Task 3.5: Attention mechanism comparison [REVIEW]

What. At the best-performing scale from Task 3.4, compare the three attention variants:

1. Option A: Integer attention with fixed scaling (already implemented).
2. Option B: Hadamard attention (implement from the equations in the revised thesis §4.5).

3. Option C: Linear attention with binary feature maps (implement binary LSH kernel).

Measure perplexity, training stability, and inference arithmetic profile (what fraction of operations are binary).

Estimated effort. 1–2 weeks.

Addresses. Moderate Concern 8 (attention mechanisms underdeveloped).

4.6 Phase 3 Deliverables

1. Complete ablation results at small scale (figures, tables).
2. Perplexity results at $d \in \{256, 512, 1024\}$ against all baselines.
3. Attention mechanism comparison.
4. Spectral diagnostic plots over training for all configurations.
5. Determination of whether "strong," "moderate," or "diagnostic" success criterion is met.

5 Phase 4: Extended Experiments and Analysis (Weeks 21–30)

Goal: Run the remaining ablations, perform the zero-shot evaluation, and develop the inference pipeline analysis.

5.1 Task 4.1: Additional ablations (A7, A8) [REVIEW]

- **A7: Sinkhorn rounding sensitivity.** For Hadamard-structured layers, measure performance when soft Sinkhorn permutations are rounded to hard permutations at various points during training (early, mid, late, only at end).
- **A8: Column encoding.** Standard binary vs. Gray code vs. learned permutation.

Estimated effort. 1 week.

Addresses. Moderate Concerns 2 (column indexing) and 7 (Sinkhorn leakage).

5.2 Task 4.2: Zero-shot evaluation [THESIS]

What. Evaluate the best SBT model on downstream benchmarks via lm-evaluation-harness:

- LAMBADA (cloze completion)
- HellaSwag (commonsense reasoning)
- ARC-Easy and ARC-Challenge (science QA)
- PIQA (physical intuition)
- WinoGrande (coreference)

Compare against all baselines. This is the standard evaluation expected by the binary network community.

Estimated effort. 3–5 days (evaluation is fast; the work is in setting up lm-eval-harness integration).

5.3 Task 4.3: Inference pipeline analysis [REVIEW]

What. Produce a complete accounting of the arithmetic operations in the SBT inference pipeline:

- For each layer: count binary operations (additions/subtractions via XNOR+popcount), integer operations, and any remaining floating-point operations.
- Identify exactly where floating-point leaks in: softmax, normalization, embeddings.
- Quantify: what fraction of total operations are strictly binary? How does this compare to BitNet?
- Prototype a pure-integer inference path (if possible) using fixed-point arithmetic for the non-binary components.

Estimated effort. 1–2 weeks.

Addresses. Major Concern 3 (scope of "purely binary" claim), Moderate Concern 6 (scaling arithmetic), Suggestion 3 (tighten claims or qualify).

5.4 Task 4.4: Multi-layer composition analysis [REVIEW]

What. Empirically study how spectral structure evolves through the network:

- At each layer, compute the effective spectral degree of the *activations* (after binarization), not just the weights.
- Track noise stability Stab_ρ across layers. Does it decay with depth, and under what conditions? The composition identity in §2.8 of the thesis provides a framework for interpreting the results.
- Monitor the mutual information diagnostics from the thesis (§4.1): the empirical output entropy $H(\mathrm{sgn}(W_\ell y_{\ell-1}))$ per layer (estimated over minibatches) and the per-row total influence $\mathbf{I}[f_i]$. If $H(\mathrm{sgn}(Wx))$ drops below $\log_2 V$ at any layer, the sign bottleneck is binding.
- Correlate layer-wise spectral properties with gradient magnitudes.

Estimated effort. 1 week.

Addresses. Major Concern 4 (activation precision hand-waved), the new Open Problem 3 (multi-layer composition).

5.5 Phase 4 Deliverables

1. Zero-shot benchmark results table.
2. Complete inference arithmetic breakdown (table + discussion).
3. Layer-wise spectral composition analysis (figures).
4. All remaining ablation results (A7, A8).

6 Phase 5: Writing and Defense (Weeks 31–40)

6.1 Task 5.1: Thesis writing

Expand the proposal into a full thesis. The revised proposal already contains substantial text for Chapters 1–4. The remaining writing:

- **Chapter 5: Pilot Study Results.** Present the spectral concentration findings from Phase 1 with figures and analysis.
- **Chapter 6: Experimental Results.** All results from Phases 3–4, organized by ablation and scale.

- **Chapter 7: Analysis and Discussion.** Inference pipeline analysis, multi-layer composition study, and honest assessment of which success criterion was met.
- **Chapter 8: Conclusion and Future Work.** What was learned, what remains open, and concrete next steps.

Estimated effort. 4–6 weeks.

6.2 Task 5.2: Open-source release

Package and release:

- `spectral-binary-nn`: Python package with `SpectralLinear`, `HadamardLinear`, `SpectralDiagnostics`, and `PilotAnalysis`.
- Pretrained SBT checkpoints at each scale.
- Reproducibility scripts for all experiments.

Estimated effort. 1 week (code cleanup, documentation, tests).

7 Summary: Review Concern Traceability

The following table maps each committee concern to where it is addressed in both the revised thesis and this work plan.

Committee Concern	Revised Thesis	Work Plan Task
Major 1: Core analogy may not hold	§2.4 expanded remark; new §5 (Pilot Study)	Tasks 1.1, 1.2 (pilot study)
Major 2: Column-indexing arbitrary	New §4.2.2; invariance argument	Tasks 1.3, 4.1 (ablation A8)
Major 3: "Purely binary" oversold	New "Scope and claims" paragraph in §1.3; precise three-part claim	Task 4.3 (inference analysis)
Major 4: Activation precision hand-waved	New §2.8 (multi-layer composition); composition identity (no longer a numbered proposition)	Task 4.4 (composition analysis); Ablation A5

Committee Concern	**Revised Thesis**	**Work Plan Task**
Major 5: Missing theory	New §3 (Theoretical Contributions): Prop. 3.1, Prop. 3.2, Conjectures 3.5–3.6	Task 2.1 (strengthen theory)
Moderate 6: Scaling arithmetic	Prop. 3.2 & Remark 3.3 (normalization propagation)	Task 4.3 (inference pipeline)
Moderate 7: Sinkhorn leakage	Acknowledged in §4.3; new Ablation A7	Tasks 2.3, 4.1 (ablation A7)
Moderate 8: Attention underdeveloped	§4.4 expanded with equations, ranking, and rationale	Task 3.5 (comparison)
Moderate 9: Bibliography issues	WHNO cite key fixed (`cavallazzi2025whno`); Merity date corrected; Alman arXiv corrected; foundational BNN references added	(Done in revision)
Suggestion: Pilot experiment	New §5 with full protocol	Tasks 1.1–1.2
Suggestion: Theory section	New §3	Task 2.1
Suggestion: Tighten binary claims	§1.3 "Scope and claims"	Task 4.3
Suggestion: Commit to attention	§4.4 with ranked options	Task 3.5
Suggestion: Column-indexing	§4.2.2	Tasks 1.3, 4.1
Suggestion: Toy model	New §4.7 (Majority function)	Task 1.4

8 Timeline Overview

Phase	Weeks	Key Milestone	Gate?
1: Pilot & Foundation	1–4	Spectral concentration results	Yes
2: Theory & Core Impl.	5–10	Defense-ready package	—
— Proposal Defense —			
3: Core Experiments	11–20	SBT results vs. baselines	Yes
4: Extended & Analysis	21–30	Zero-shot eval; inference analysis	—
5: Writing & Defense	31–40	Thesis submission	—

Decision gates.

- **End of Phase 1:** If no spectral concentration observed, pivot to Hadamard-structured approach and/or reframe contributions. Do not proceed to Phase 3 with the row-wise spectral parameterization.
- **End of Phase 3 (Task 3.2):** If SBT at small scale cannot outperform random binary weights, diagnose and fix before scaling up.

9 Resource Requirements

Phase	Compute	Notes
1	CPU only	Spectral analysis; toy models
2	1 GPU (A100 or equiv.)	STE baseline training
3	2–4 GPUs	Multiple training runs at $d \leq 1024$
4	1–2 GPUs	Evaluation; analysis
5	Minimal	Writing

Total estimated GPU-hours: 2,000–5,000 (modest by current standards; all models are small by design).

10 Immediate Next Actions (This Week)

1. **Monday–Tuesday:** Implement the FWHT and spectral analysis functions (Task 1.1).
2. **Wednesday:** Download BitNet b1.58 2B4T weights; binarize GPT-2 Small weights. Run full spectral analysis on both (Task 1.2).
3. **Thursday:** Column-indexing sensitivity tests (Task 1.3). Generate all pilot study figures.

4. **Friday:** Implement and run the Majority function toy model (Task 1.4). Review Phase 1 results and make the go/no-go decision.

If Phase 1 results are positive, begin Task 2.1 (theory) and Task 2.2 (SpectralLinear implementation) the following Monday.